

Investigate the stability of the torque (issue #18)

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1 Short summary

This report checks the torque sampling used for `motor1`. With the current `PointPer = 101` and $N_{\text{per}} = 1/6$, Ansys returns only 18 torque values. The aim is to check whether this is sufficient for `TorAvg` and `TorRippleRms`, and to find a practical value of `PointPer` for stable results.

2 Method

For `motor1`, the transient settings are

$$\text{StopTime} = \frac{N_{\text{per}}}{f}, \quad \text{TimeStep} = \frac{1}{f(\text{PointPer} - 1)}.$$

Here, `PointPer` controls the number of sampling points per period, while $N_{\text{per}} = 1/6$ of a period is simulated.

Two fixed Pareto optimal rotor designs were tested, one from `HacklGenerator_SixLambdas` and one from `HacklGenerator_3BrokenLines`. For each design, `PointPer` was varied from 51 to 1001 in steps of 50, corresponding to 10 to 168 returned torque samples. All other settings were kept fixed. `TorAvg` and `TorRippleRms` were evaluated with `analyze_results()`. The sweep is implemented in `scripts/sensitivity_torque/sensitivity_torque.py`.

3 Results

Figure 1 shows the convergence of these two torque quantities with increasing `PointPer`.

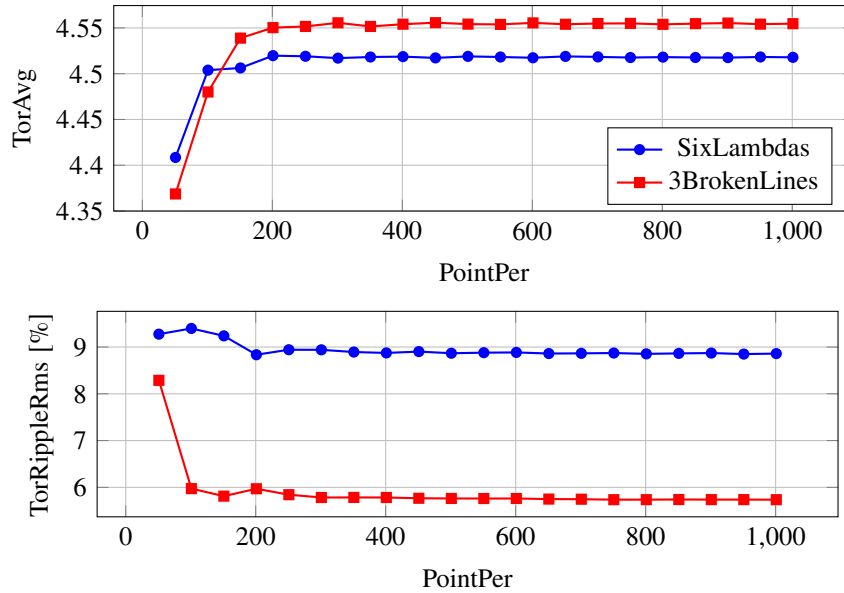


Figure 1: Convergence of mean torque and torque ripple with increasing `PointPer`.

4 Stability analysis

The results are stable at large `PointPer`, with only small numerical fluctuations. Therefore, the mean of the five largest `PointPer` cases (`PointPer = 801–1001`) was used as the reference value. The relative error is

$$\varepsilon_q = \frac{|q - q_{\text{ref}}|}{|q_{\text{ref}}|} \times 100\%$$

At the current $\text{PointPer} = 101$, the errors in TorAvg and TorRippleRms are 0.31% and 6.07% for SixLambdas , and 1.64% and 4.10% for 3BrokenLines . The ripple is more sensitive to low sampling density.

Both quantities remain within 1% of the reference from $\text{PointPer} = 301$ onward. For a tighter 0.5% limit, $\text{PointPer} = 501$ can be used.

5 Recommendation

For motor1 , $\text{PointPer} = 301$ is recommended for stable mean torque and torque ripple. If an error below 0.5% is required, $\text{PointPer} = 501$ can be used instead. The current $\text{PointPer} = 101$ is not sufficient for accurate torque ripple.

These results apply to motor1 and the two designs tested here.